

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE Supplementary Examination Summer – 2023 Course: B. Tech. Branch : Mechanical Engineering Semester :VII Subject Code & Name: BTMEC703 & Manufacturing Processes - III Max Marks: 60 Date: Duration: 3 Hr.			
Instructions to the Students: 1. All the questions are compulsory. 2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question. 3. Use of non-programmable scientific calculators is allowed. 4. Assume suitable data wherever necessary and mention it clearly.			
		(Level/CO)	Marks
Q. 1	Solve Any Two of the following.		12
A)	Explain with neat sketches the interpolations performed in CNC machining operations.	(CO1)	6
B)	Distinguish with a neat sketch between P-T-P and continuous path motion control systems in CNC Machines.	(CO1)	6
C)	Explain with a neat sketch the mechanism of feedback system employing rotary encoder in CNC Machines.	(CO1)	6
Q.2	Solve Any Two of the following.		12
A)	What do you understand by automatic tool changer? Describe in brief any two types with the help of sketches.	(CO2)	6
B)	Illustrate the program starting declarations so that the CNC machine will execute the desired machining operations.	(CO2)	6
C)	Write down programming codes for performing following functions on CNC Machines: Absolute programming, Incremental Programming, Rapid tool positioning, Circular interpolation(CCW), Spindle Start(Clockwise) and Coolant On.	(CO2)	6
Q. 3	Solve Any Two of the following.		12
A)	Explain with a neat sketch the construction and working principle of LBM process.	(CO3)	6
B)	What is the principle of EDM process? Describe advantages, limitations & applications of EDM process over conventional machining processes.	(CO3)	6
C)	Explain with a neat sketch the electrochemical machining process. State its advantages over the traditional machining processes.	(CO3)	6

Q.4	Solve Any Two of the following.		12
A)	What are the properties obtained by electroplating process? Explain with a neat sketch the principle of electroplating.	(CO4)	6
B)	Describe with suitable sketch physical vapour deposition process. State important applications of PVD.	(CO4)	6
C)	Describe in brief the steps in micro-fabrication process LIGA. Describe advantages of LIGA over other processes.	(CO6)	6
Q. 5	Solve Any Two of the following.		12
A)	State the advantages of Rapid Prototyping processes over traditional manufacturing processes. Distinguish between additive and subtractive manufacturing processes.	(CO5)	6
B)	Describe Solid Ground Curing (SGC) process. State its advantages and limitations.	(CO5)	6
C)	Differentiate between surface and bulk micromachining of MEMS.	(CO6)	6
	*** End ***		

The grid and the borders of the table will be hidden before final printing.